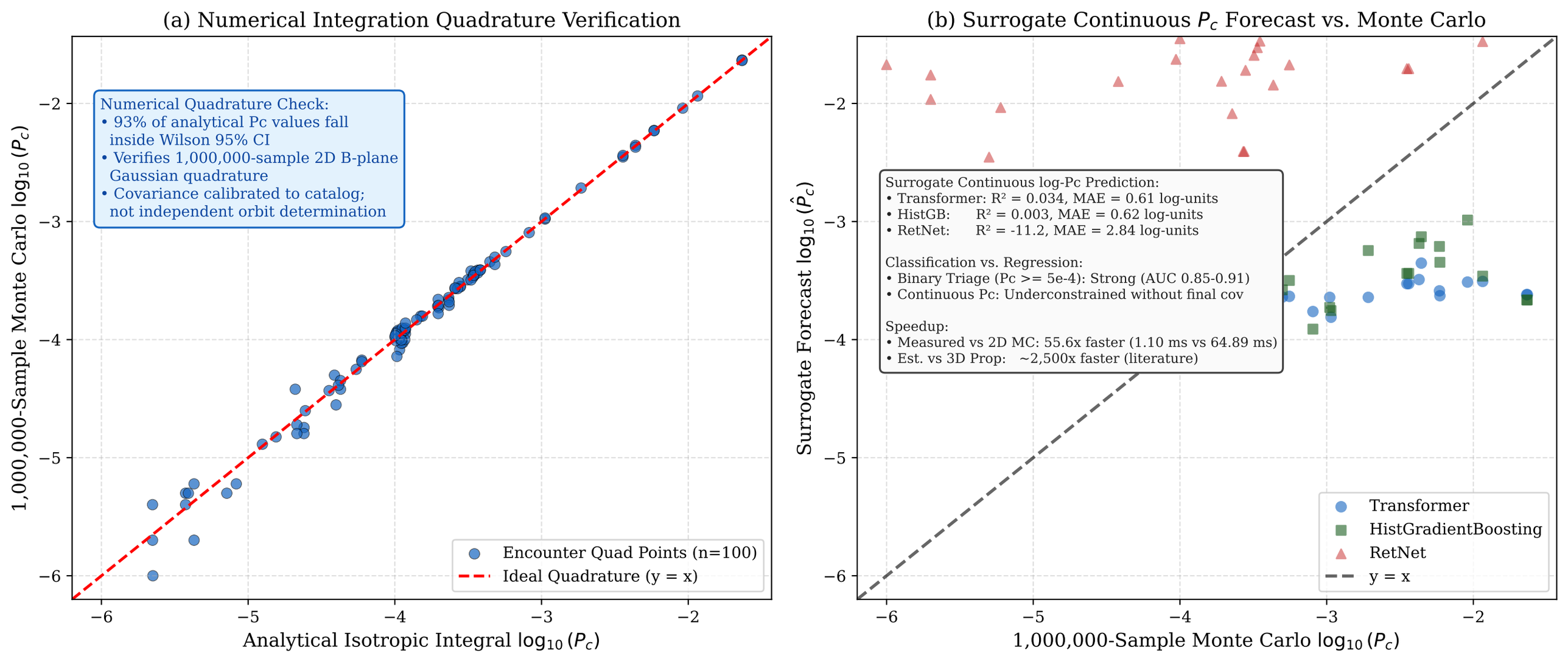
(a) Numerical Integration Quadrature Verification
Numerical Quadrature Check:
• 93% of analytical Pc values fall inside Wilson 95% CI
• Verifies 1,000,000-sample 2D B-plane Gaussian quadrature
• Covariance calibrated to catalog; not independent orbit determination
Encounter Quad Points (n=100)
Ideal Quadrature (y = x)
1,000,000-Sample Monte Carlo log10 (Pc)
Analytical Isotropic Integral log10 (Pc)
(b) Surrogate Continuous Pc Forecast vs. Monte Carlo
Surrogate Continuous log-Pc Prediction:
• Transformer: R² = 0.034, MAE = 0.61 log-units
• HistGB: R² = 0.003, MAE = 0.62 log-units
• RetNet: R² = -11.2, MAE = 2.84 log-units
Classification vs. Regression:
• Binary Triage (Pc >= 5e-4): Strong (AUC 0.85-0.91)
• Continuous Pc: Underconstrained without final cov
Speedup:
• Measured vs 2D MC: 55.6x faster (1.10 ms vs 64.89 ms)
• Est. vs 3D Prop: ~2,500x faster (literature)
Transformer
HistGradientBoosting
RetNet
y = x
Surrogate Forecast log10 (P̂c)
1,000,000-Sample Monte Carlo log10 (Pc)